

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Revision Test 2026

BIOLOGY — SSC-I (9th Grade) — ANSWER KEY (Version 15)

Syllabus: Chapter 1 - The Science of Biology, Chapter 2 - Biodiversity, Chapter 3 - The Cell, Chapter 4 - Cell Cycle, Chapter 5 - Tissues, Organs and Organ Systems, Chapter 6 - Molecular Biology, Chapter 7 - Metabolism (Sections 7.1 to 7.6 only)

Total Marks: 43

Time Allowed: 2 Hours

Version 15

SECTION A — Answers to Multiple Choice Questions (24 × 1 = 24 Marks)

Q1. The study of the development of an organism from a fertilized egg is called:

Correct answer: **C) embryology** 1 mark

Embryology is the study of the development of an organism from a fertilized egg. Paleontology is the study of the history of life on Earth based on fossils, histology is the microscopic study of tissues, and taxonomy is the classification and naming of organisms.

Source in her textbook: Ch 1, §1.2 Sub-fields of Biology — SLO 4 (sub-fields of biology with examples)

Q2. A statement that describes an observable occurrence in nature which appears to always be true is called a:

Correct answer: **D) scientific law** 1 mark

A **scientific law** is a statement that describes an observable occurrence in nature that appears to always be true. A theory that has been verified and appears to have wide application may be elevated to the status of a law, for example **Mendel's law of inheritance**. A hypothesis is only a tentative, testable answer.

Source in her textbook: Ch 1, §1.7 Hypothesis, Theory and Law; Summary point 13 — SLO 9 (evaluate hypothesis, theory and law)

Q3. The collection of biological data through observations and experiments, and analysing it according to statistical rules, is called biostatistics. It is also known as:

Correct answer: **A) biometry** 1 mark

Biostatistics is the collection of biological data or facts through observations and experiments and analysing them according to statistical rules. The textbook states that it **is also called biometry**.

Source in her textbook: Ch 1, Table 1.1 Relationship of Biology with other Sciences — SLO 5 (biology connects with other natural sciences)

Q4. A group of related genera makes up a:

Correct answer: **B) family** 1 mark

The taxonomic ladder runs domain, kingdom, phylum (division), class, order, family, genus, species. A **family** is a group of related genera, an order is a group of related families and a class is a group of related orders.

Source in her textbook: Ch 2, §2.5 Taxonomic Ranks of Classification — SLO 5 (list the taxonomic ranks of classification)

Q5. The classification of living organisms into three domains is based on differences in the sequence of nucleotides in the:

Correct answer: **C) rRNA of the cell** 1 mark

Classification into three domains is based on differences in the sequence of nucleotides in the **rRNA (ribosomal ribonucleic acid)** of the cell, on the cell's membrane lipid structure, and on its sensitivity to antibiotics.

Source in her textbook: Ch 2, §2.4 Classification — The Three Domains System — SLO 4 (three distinct domains)

Q6. A group of similar organisms whose individuals are capable of interbreeding and producing fertile offspring is called a:

Correct answer: **A) species** 1 mark

A **species** is a group of similar organisms, the individuals of which are capable of interbreeding or exchanging genes among themselves and producing **fertile offspring**. A mule is not a species because it is infertile.

Source in her textbook: Ch 2, §2.6 Species — SLO 6 (define species)

Q7. According to the fluid mosaic model, the protein molecules of the cell membrane float like icebergs in a sea-like fluid of:

Correct answer: **D) phospholipids** 1 mark

The fluid mosaic model postulates that the cell membrane consists of a **double layer of phospholipids** in which proteins are incorporated in a mosaic fashion. The textbook states that protein molecules float like icebergs in a sea-like fluid of phospholipids.

Source in her textbook: Ch 3, §3.1.1 Cell membrane / plasma membrane — SLO 3 (sketch subcellular organelles including the cell membrane and outline their roles)

Q8. In a neuron, the cytoplasmic fibres that conduct nerve impulses *towards* the cell body are called:

Correct answer: **B) dendrites** 1 mark

A neuron has a cell body and two types of cytoplasmic fibres. **Dendrites** conduct nerve impulses to the cell body, while **axons** conduct messages away from the cell body.

Source in her textbook: Ch 3, §3.2 Cell Specialization (Neurons), Fig 3.20 — SLO 5 (identify different types of cells including neurons)

Q9. The most basic stem cell, which has the ability to produce all the kinds of cell found in an organism, is the:

Correct answer: **A) zygote** 1 mark

A cell which gives rise to cells of other types is called a stem cell. The textbook states that the **zygote** is the very basic stem cell, since it has the ability to produce all kinds of cell of an organism. Neurons, red blood cells and liver cells are specialised cells.

Source in her textbook: Ch 3, §3.4 Stem cells — SLO 10 (define stem cells as unspecialized cells)

Q10. The cell is regarded as the basic unit of life because it is the:

Correct answer: **C) smallest unit of living material** 1 mark

The textbook summary states that the cell is considered the basic unit of life because it is the **smallest unit of living material**. All the functions performed by a living organism are also performed at the cell level.

Source in her textbook: Ch 3, §3.1 The Cell; Summary point 1 — SLO 1 (describe cell as the basic unit of life)

Q11. In meiosis a diploid cell divides:

Correct answer: **D) twice, producing four haploid cells** 1 mark

A diploid cell undergoing meiosis completes **two successive cell divisions** (meiosis I and meiosis II) to give rise to **four haploid cells**, which later form gametes. Meiosis I is the reduction division that halves the chromosome number.

Source in her textbook: Ch 4, §4.5 Meiosis; Summary point 10; Table 5.7 — SLO 2 and SLO 3 (explain meiosis; compare mitosis and meiosis)

Q12. Which of the following takes place in meiosis but never in mitosis?

Correct answer: **B) Homologous chromosomes pair with each other** 1 mark

In the textbook's comparison table, all chromosomes remain independent of each other in mitosis, whereas in meiosis **homologous chromosomes pair with each other** (synapsis). Spindle formation, condensation of chromatin and disappearance of the nuclear envelope happen in both.

Source in her textbook: Ch 4, Table 5.7 Comparison of mitosis and meiosis — SLO 3 (compare the process of mitosis and meiosis)

Q13. The replacement of worn-out cells and the healing of wounds depend on:

Correct answer: **C) mitosis** 1 mark

Cell replacement and wound healing are listed among the significances of **mitosis**. Epithelial cells of the skin, digestive tract and respiratory tract die off regularly and are replaced by identical cells produced by mitosis.

Source in her textbook: Ch 4, §4.3.1 Significance of Mitosis — SLO 4 (outline the significance of mitosis and meiosis)

Q14. In an animal cell, cytokinesis is completed by the inward growth of a:

Correct answer: **A) cleavage furrow** 1 mark

In an animal cell the cell membrane begins to invaginate in the equator region, forming a **cleavage furrow** which continues to grow inward until the membranes join and separate the two daughter cells. The phragmoplast and middle lamella belong to plant-cell cytokinesis.

Source in her textbook: Ch 4, §4.3 Mitosis (Cytokinesis), Fig 4.4 — SLO 2 (stages of mitosis by use of sketch and diagrams)

Q15. Which of the following is **NOT** one of the four main types of animal tissue?

Correct answer: **D) Vascular** 1 mark

Animal tissues differentiate into four main types: **epithelial, connective, muscular and nervous**. Dermal, ground and **vascular** are the three main types of *plant* tissue.

Source in her textbook: Ch 5, §5.3 Plant Tissues; Summary points 8 and 9 — SLO 3 (enlist the different types of tissues with examples from animals and plants)

Q16. The inside of the stomach is lined by cells that secrete mucus, hydrochloric acid and the enzyme pepsin. These cells are:

Correct answer: **B) epithelial cells** 1 mark

The layered walls of the stomach show how tissues form an organ. The inside of the stomach is lined by **epithelial cells** which secrete mucus, hydrochloric acid and pepsin. Around this layer are connective tissue and smooth muscle, along with glands, blood vessels and neurons.

Source in her textbook: Ch 5, §5.1.1 Organs (stomach), Fig 5.2 — SLO 4 (how tissues come together to form the stomach organ)

Q17. The waxy substance that covers the upper epidermis of a leaf and forms the cuticle is:

Correct answer: **A) cutin** 1 mark

A waxy substance called **cutin**, which forms the cuticle, covers the upper epidermis of the leaf. Its function is to protect the tissues and prevent loss of water. Lignin cements cellulose in secondary cell walls, pectin forms the middle lamella and chitin makes fungal cell walls.

Source in her textbook: Ch 5, §5.3.2 Structure and Functions of Cells of a Leaf — SLO 8 (types of tissues that form the leaf)

Q18. Every amino acid has a central α -carbon carrying a hydrogen atom, an amino group and a carboxyl group. One amino acid differs from another because of its:

Correct answer: **C) alkyl (R) group** 1 mark

Four different groups are attached to the α -carbon of an amino acid: a hydrogen atom ($-H$), an amino group ($-NH_2$), a carboxyl group ($-COOH$) and an alkyl group ($-R$). The textbook states that each amino acid has a different **-R group**, which determines its chemical properties.

Source in her textbook: Ch 6, §6.3.2 Amino acids, Fig 6.2 — SLO 3 (structure of proteins with structure of amino acids)

Q19. Lipids are insoluble in water but soluble in organic solvents such as alcohol and ether. Because of this they are described as:

Correct answer: **D) hydrophobic** 1 mark

Lipids are a heterogeneous group of organic compounds which are all **hydrophobic** in nature: they are insoluble in water but soluble in organic solvents such as alcohol and ether.

Source in her textbook: Ch 6, §6.4 Lipids — SLO 4 (outline the structure, function and sources of lipids)

Q20. Monosaccharides are named according to the number of carbon atoms they contain. A monosaccharide with six carbon atoms, such as glucose ($C_6H_{12}O_6$), is called a:

Correct answer: **B) hexose** 1 mark

Monosaccharides are named on the basis of the number of carbon atoms in their structure: trioses (3C), tetroses (4C), pentoses (5C) and **hexoses (6C)**. Glucose, $C_6H_{12}O_6$, is a familiar hexose; ribose is a pentose.

Source in her textbook: Ch 6, §6.6.1 Monosaccharides — SLO 6 (identify carbohydrates as monosaccharides, disaccharides and polysaccharides)

Q21. The type of RNA that transfers amino acids to the ribosome during protein synthesis is:

Correct answer: **A) tRNA** 1 mark

There are three types of RNA. mRNA carries the message of DNA from the nucleus to the cytoplasm, **tRNA (transfer RNA)** transfers amino acids to the ribosomes during protein synthesis, and rRNA combines with ribosomal protein to form the subunits of ribosomes.

Source in her textbook: Ch 6, §6.10.1 Types and functions of RNA — SLO 10 (function of RNA in converting hereditary information into proteins)

Q22. The study of the structure, type and role of enzymes is called:

Correct answer: **C) enzymology** 1 mark

The textbook states that the study of the structure, type and role of enzymes is called **enzymology**. Metabolism is the sum of all the chemical reactions of a cell, physiology is the study of the functions of organs, and cytology is the study of the cell.

Source in her textbook: Ch 7, §7.2 Enzymes and their Characteristics — SLO 2 (define enzymes and describe their characteristics)

Q23. The condensation of glucose molecules to form glycogen in muscle and liver cells is an example of:

Correct answer: **B) anabolism** 1 mark

Reactions which build complex molecules from simpler molecules are called anabolic reactions or **anabolism**. The textbook gives the condensation of glucose molecules to form glycogen in muscle and liver cells as an example. Digestion is a catabolic process.

Source in her textbook: Ch 7, §7.1 Metabolism and its Types — SLO 1 (define metabolism, catabolism and anabolism with examples)

Q24. ATP is converted to ADP and inorganic phosphate by the process of:

Correct answer: **D) hydrolysis** 1 mark

In the ATP-ADP cycle, ATP is converted to ADP and inorganic phosphate by **hydrolysis**: the third phosphate group separates from ATP and remains in the cell in inorganic form. ADP and phosphate are converted back to ATP by **condensation**, using energy from catabolism or from light energy absorbed by plants.

Source in her textbook: Ch 7, §7.6.1 ATP - ADP Cycle, Fig 7.7 — SLO 6 (discuss the role of ATP as energy currency)

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	C	D	A	B	C	A	D	B	A	C	D	B
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	C	A	D	B	A	C	D	B	A	C	B	D

SECTION B — Answers to Short Questions (3 × 3 = 9 Marks)

Q.2(i) Table 2.1 of your textbook gives a simple classification of the pea plant, *Pisum sativum*. Write out its taxonomic ranks from Kingdom down to Species, giving the name your textbook lists at each rank. 3 marks

Taxon (rank)	Name for the pea
Kingdom	Plantae
Division (phylum in plants)	Magnoliophyta
Class	Magniopsida
Order	Fabales
Family	Fabaceae
Genus	<i>Pisum</i>
Species	<i>Pisum sativum</i>

Marking: **1 mark** Kingdom, Division and Class all correct. **1 mark** Order and Family both correct. **1 mark** Genus and Species both correct, with the genus name capitalised and the species name in small letters.

Note: plants use **Division** where animals use **Phylum**. Accept *Magnoliopsida* for *Magniopsida*, which is the spelling printed in her book.

Source in her textbook: Ch 2, Table 2.1 — SLO 5 (taxonomic ranks) and SLO 7 (binomial nomenclature).

Q.2(ii) An animal cell has no cell wall. State THREE advantages that an animal cell gains because it lacks a cell wall, as given in your textbook. 3 marks

Any **three** of the following, one mark each:

- Flexibility and movement.** Lacking a cell wall makes animal cells very flexible, so animal cells (and animals) can move from place to place, find shelter, reach better feeding fields and move to suitable environmental conditions. **1 mark**
- Faster division.** Because it is not enclosed by a rigid structure, an animal cell can divide and reproduce at a faster rate than a plant cell. **1 mark**
- Support comes from the body as a whole.** The supportive structure of an animal does not depend on a cell wall but on the collective arrangement and organisation of tissues, organs and skeletal systems. **1 mark**
- Different transport routes.** Since animal cells lack a cell wall, the transport of fluids, nutrients and gases occurs through different structures rather than through xylem and phloem. **1 mark**

Do not credit the two disadvantages from the same table: an animal cell cannot withstand high osmotic pressure or store large volumes of water, and it cannot become turgid to provide support.

Source in her textbook: Ch 3, §3.1.2 Structural advantages of animal and plant cell (comparison table) — SLO 4.

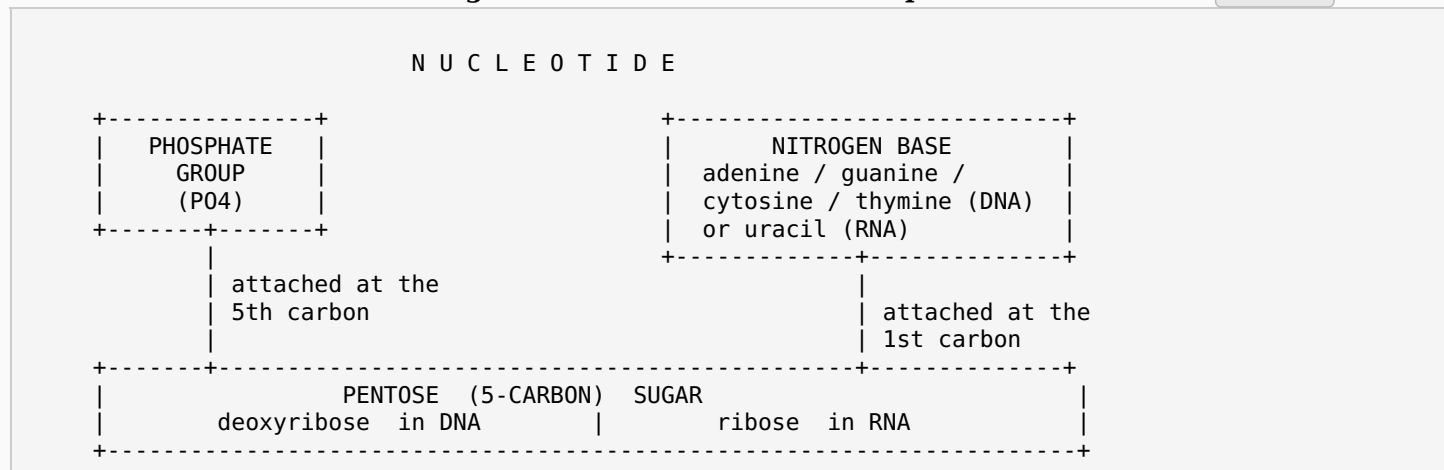
Q.2(iii) For carbohydrates, lipids and DNA, state ONE location inside the cell and ONE main role of each (Table 6.1). 3 marks

Biomolecule	Location in the cell	Main role
Carbohydrates (glucose, sucrose, starch, cellulose)	Cell wall, cell membrane, smooth endoplasmic reticulum, Golgi bodies	Used as food and as an energy source; component of plant and fungal cell walls; cell-surface markers; the 5-carbon sugar of DNA and RNA
Lipids (fats, phospholipids, steroids)	Cell membrane, smooth endoplasmic reticulum	Provide cells with long-term energy; component of cell membranes
DNA	Nucleus (in the chromosomes)	Component of chromosomes; stores and passes on genetic information

Marking: **1 mark** for each row, awarded only when **both** the location and a role are correct. Any one correct role from the textbook column is enough.

Source in her textbook: Ch 6, Table 6.1 Biomolecules, their location in the cell and roles — SLO 2.

Q.3(i)(a) Draw a well-labelled diagram of a nucleotide, showing its three components and stating to which carbon atom of the sugar each of the other two components is attached. 3 marks



Each nucleotide is made up of **three** components:

1. **Pentose (5-carbon) sugar** — deoxyribose in DNA, ribose in RNA. **1 mark**
2. **Nitrogen base**, a nitrogen-containing ring structure, attached at the **first carbon** of the pentose sugar. **1 mark**
3. **Phosphate group**, attached at the **fifth carbon** of the pentose sugar. **1 mark**

Marking note: award the mark for each of the three parts correctly named **and** correctly placed. Deduct nothing for the style of drawing; a clear labelled block diagram is acceptable. If she names all three parts but gives no attachment points, award 2 of the 3 marks.

Q.3(i)(b) Name the bond that holds the two strands of DNA together, state how many lie between adenine and thymine and how many between cytosine and guanine, and state what the backbone of the DNA molecule is made of. 2 marks

Both strands of DNA are held together by weak **hydrogen bonds**. There are **two** hydrogen bonds between adenine and thymine and **three** hydrogen bonds between cytosine and guanine. **1 mark**

The backbone (uprights) of the molecule is composed of the **sugar and phosphate** parts of the nucleotides, while the **bases form the rungs** of the ladder. **1 mark**

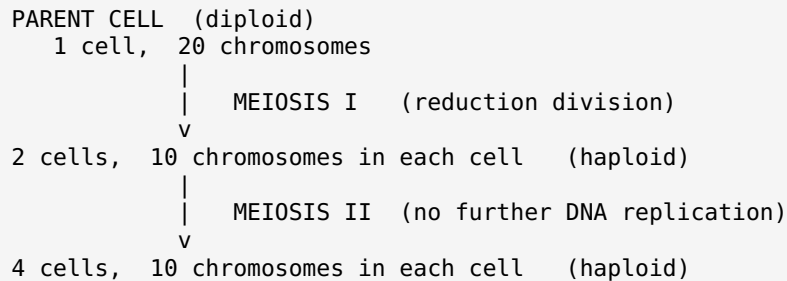
Source in her textbook: Ch 6, §6.7.1 Structure of Nucleotide (Fig 6.11) and §6.7.2 Structure of DNA, points 3, 5 and 6 — SLO 7 and SLO 8.

Q.3(ii) A cell in the ovary of a plant contains 20 chromosomes and undergoes meiosis. 5 marks

(a) The two successive divisions of meiosis are **meiosis I** and **meiosis II**. 1 mark

Meiosis I is the **reduction division**: it is the division that reduces the number of chromosomes to half, because the homologous chromosomes of each pair are separated and move to opposite poles. Meiosis II is not a reduction division; it simply separates the two chromatids of each chromosome, and no DNA replication takes place in interphase II. 1 mark

(b) Flow chart 2 marks



1 mark for 2 cells with 10 chromosomes each at the end of meiosis I. 1 mark for 4 cells with 10 chromosomes each at the end of meiosis II.

Common error to watch for: writing 4 cells of 20 chromosomes, or 2 cells at the end of meiosis II. Meiosis divides **twice** and always ends with **four** cells, each with **half** the parent chromosome number.

(c) The cells produced at the end of meiosis are **haploid** (they carry half the parent chromosome number, one member of each homologous pair). They go on to form the **gametes**, that is the sex cells; in plants, meiosis takes place during spore formation. 1 mark

Source in her textbook: Ch 4, §4.5 Meiosis, §4.4.1 Significance of meiosis, Summary points 10 and 12, Table 5.7 — SLO 2, SLO 3 and SLO 4.

— END OF ANSWER KEY —